

Seasonal Calendar of Mirjapur Village (M.P.)

Based on the data collected on 23.08.2025 through survey and discussions.

GROUP REPORT

Submitted in partial fulfilment of the requirements for the completion of courses:

IE 301/401/601 Introduction to Entrepreneurship

&

RDT 301/401/601 Design Thinking for Rural Application



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Introduction

We went to **Mirjapur** village, Madhya Pradesh, on 23rd August 2024. To make the seasonal calendar, we conducted a questionnaire session with villagers and farmers of Mirjapur. It was made sure that the survey was conducted as a discussion and not as an Interview/question-answer examination; This approach ensured that the villagers were open and shared their problems and concerns with us. We first divided ourselves into 3 Groups of 4, 3 and 3 members, and the teams conducted a survey in different village areas.



Map 1: team division and village traversing followed for survey and Data collection.

(Areal Data Source: Airbus CNES, Airbus Maxar Technologies via Google Maps. Right to reproduce this map under Google's fair use policy for education.)

Group Division

The group division ensured that we bring in different ideas and different fields of expertise into play, like one team with specialisation in CRDT and Biosciences taking diseases, etc., into account, a PhD group of EE, ME, and MEMS would bring in innovation on the desk, and a B.Tech. student' group may bring a fresh perspective. The division was as follows:

Group A (4 Members)

Rudrajit Mandal	2501171016	PhD (BSBE)
Rohit Saha	2501171004	PhD (BSBE)
Riyanshi Verma	2501171005	PhD (BSBE)
Buddhodev Ghosh	2501192002	PhD (CRDT)

Group B (3 Members)

Dhruv Jain	220005018	B. Tech. (MEMS)
Kanika Upadhyay	220005028	B. Tech. (MEMS)
Lokery Kiranmayi	220005033	B. Tech. (MEMS)

Group C (3 Members)

Gaurav Rajput	2302105018	PhD (MEMS)
Rahul Kushwah	2302102015	PhD (EE)
Himanshu Sharma	2302103018	PhD (ME)

We visited various households for the primary data collection. According to the data collected from the villagers of Mirjapur, we have made a seasonal calendar of the village.

Most villagers were hesitant initially, but as they opened up, they cooperated well and shared with us their experiences and the information for the survey.

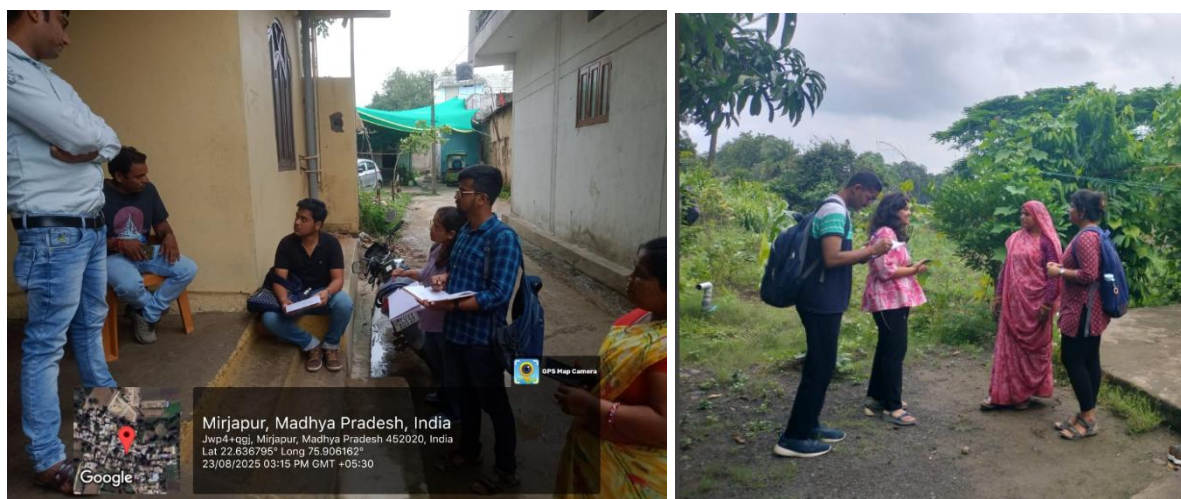


Fig. 1 a & b - Discussion with the village farmers and primary data collection.



Fig 1c (Left): Interaction with a lady associated with the labour related to farming

Fig 1d (Right): A Labourer couple who work in large farms during the prime season.



Fig 1e: Fields of Village Mirjapur as on 23 August 2025. Some fields with flower crops and other fields are left barren due to rain.

2. Seasonal Calendar

2.1. Agricultural Seasons in Mirjapur

1. **Kharif (Monsoon crops):** June – October
2. **Rabi (Winter crops):** November – March
3. **Zaid (Summer crops):** April – June

The boundaries between these seasons were found not to be very rigid and different farmers had different extremum/ boundary dates for them.

2.2 Month-Wise Seasonal Calendar

Month	Climate & Rainfall Pattern	Agricultural Activities (Crops)	Livestock & Allied Activities	Festivals / Social Events
January	Cold, dry weather	Wheat, gram, mustard, and lentils are growing	Fodder preparation, animal care	Makar Sankranti
February	Slight warming	Irrigation for wheat, mustard	Dairy & cattle grazing	Local fairs
March	Harvest of early Rabi crops	Wheat, gram harvesting begin	Fodder cutting	Holi
April	Hot, dry	Summer ploughing, pulses/vegetables	Increased water needs for livestock	Navratri/Ram Navami
May	Peak summer, water scarcity	Preparations for Kharif (field clearing)	Cattle suffer from heat stress	Local rituals for rain
June	Pre-monsoon showers begin	Sowing of Kharif crops (soybean, maize, paddy)	Grazing improves	
July	Heavy rainfall	Main sowing: soybean, maize, paddy	Milk production increases	Local village festivals
August	Rain continues	Weeding, fertiliser application	Good grazing	Independence Day
September	Rains receding	Pest management, crop growth	Collection of fodder	Ganesh Chaturthi

October	Harvest season starts	Soybean & maize harvest	Sale of produce	Dussehra, Diwali
November	Cool, dry	Wheat, gram, mustard sowing	Storage of grains	Kartik Purnima
December	Cold wave	Irrigation for Rabi crops	Stall-feeding for cattle	Marriage season begins

2.3. Additional Seasonal Aspects

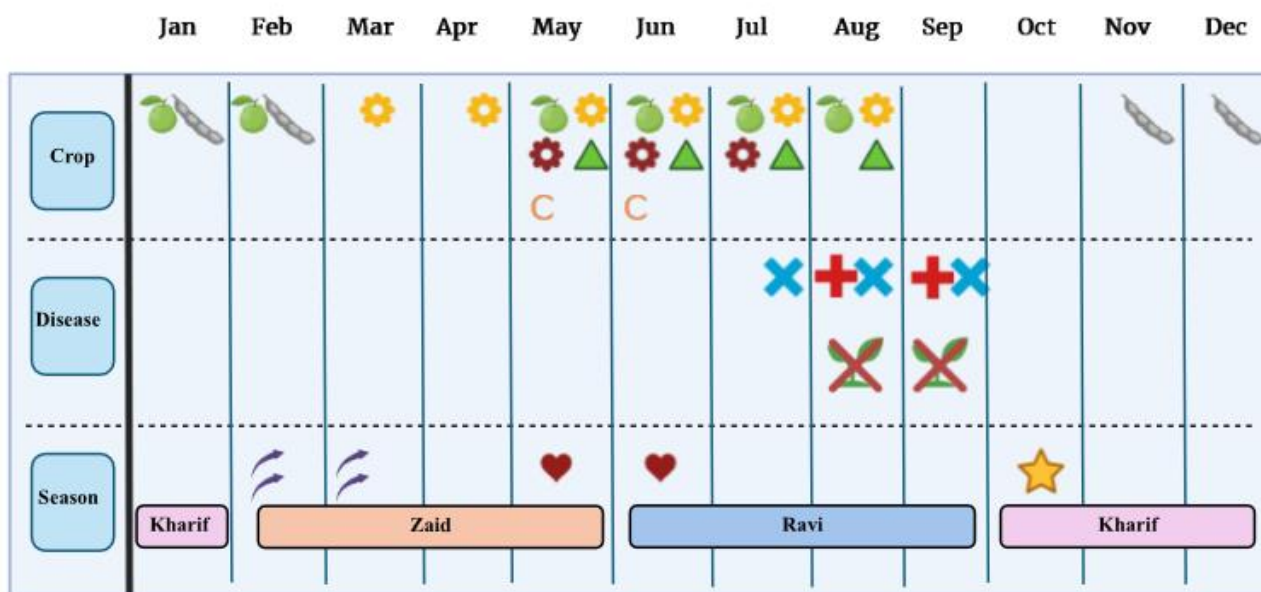
1. **Water availability:** Mainly from rainfall (July–Sept), irrigation dependency is high in Rabi.
2. **Fodder scarcity:** May–June (before monsoon).
3. **Labour demand peaks:** June–July (sowing) & March–April (harvesting).
4. **Village cultural events:** Centred around harvests and major Hindu festivals.

2.4. Seasonal Calendar Plot



Fig 2a: Seasonal Calendar developed as a rangoli pattern on the floor of the village school..

2.5. Seasonal Calendar Representative Diagram:



Legend:

- | | |
|---|---|
|  : Guava |  : Human disease |
|  : Marigold/Genda |  : Crop disease |
|  : Chrysanthemum/Sevanti |  : Crop damage |
|  : Soybean |  : Emmigration for work |
|  : Cucumber |  : Weddings |
|  : Tamarind/Imli |  : Cultural celebrations |

3. Important Inferences drawn from the data collected:

3.1 Seasonal Calendar Related:

1. Most popularly grown crops include soybean, corn, and flowers such as marigold and chrysanthemum. Interestingly, the farmers mostly wanted to be known for growing flowers in general.
2. The water source for crops is usually bore wells, and sometimes rainwater.
3. In dry seasons at the beginning of each year, a part of the population seeks jobs outside their village
4. Farmers recalled that rainfall has diminished in the last 10 years
5. A common disease among humans is malaria mainly during august and September.
6. They also use the traditional Pañcāṅga and the natural signs for the sudden rain prediction (Many of them, we talked, don't believe in the Modern Prediction systems and the News on TV and Radios).

3.2 Other agriculture-related inferences:

1. Most of the farmers surveyed sold their produce to private mandis and were unaware of the government schemes, Direct Benefit Transfer and even MSP benefits that could be availed on their produce.
2. Farmers use pesticides regularly, and yet fungal diseases can be seen in crops in the monsoon
3. Fertilisers used are mostly urea and DAP.
4. Farmers complained of insufficient support provided by the government

3.3 Major challenges faced and possible solutions

1. *Shortage of water* in the summer season for the farmers who lack borewells.
2. Crops have been getting *dehydrated* due to *adverse climate* conditions in recent years.
3. *Limited warehouse facility* to store the cultivated crop.
4. No proper place to store the cultivated crop can escalate to economic loss for farmers; for this problem preliminary solution would be to provide a *rental warehouse facility* to the farmers on a large scale, so it would benefit a greater number of farmers who are facing similar issues.
5. Many small farmers pointed out a major problem of **agricultural land shrinkage** in the village, and that residential colonies had eaten most of their agricultural lands.

Some possible solutions which could be employed:

1. For the shortage of water in the summer season, **rainwater harvesting** systems should be implemented across the village as well as the farmlands.
2. Collecting crop seed from different places, especially the crops that are *robust, resistant, and adaptive* to climate change.
3. Also, we found a need to educate the farmers about *crops from foreign* countries and how those crops can be farmed in Indian farming lands.
4. Agricultural land shrinkage is an administrative problem and will have to be dealt with by stricter policy implementation and wiser land use guidelines by the panchayat and the local administration.

To get the attention of the villagers, we had made a *rangoli* design of the seasonal calendar on the ground near the common gathering area at the entrance of Mirjapur village. It gained quite a bit of attention from villagers, including kids and the elderly, over there.



Fig. 3: Bad situation of the Village Road due to the Rainy Season



Fig. 4: Group photo with the Rangoli design of the seasonal calendar of Mirjapur village

4. Summing Up:

The Mirjapur Seasonal Calendar exercise provided us with a new way to look into the agricultural and social practices of rural Madhya Pradesh (especially near the Indore region). By visually showing yearly cycles, we started discussions on solutions and opportunities for rural entrepreneurship.